



Indian Institute of Technology Ropar
Department of Mathematics
MA302
Optimization methods with Applications
Semester - 2 of Academic Year 2025-2026

Problem Sheet - 1

1. A company produces two products A and B . Products are produced and sold on a weekly basis. The weekly production cannot exceed 20 units for product A and 30 units for product B because of limited available facilities. The company employs a total of 60 workers. Product A requires 2 man-weeks of labour, while product B requires 1 man-week of labour. The profit margin on product A is Rs. 60 and on product B is Rs. 40, respectively. Formulate this problem as a linear programming problem.
2. A firm produces two types of items, x_1 and x_2 . Due to technological and operational limitations, the production of item x_1 must always be at least one unit less than the production of item x_2 . Another operational restriction requires that the production of item x_2 cannot exceed half the production of item x_1 by more than two units. The firm earns a profit of one unit on each unit of item x_1 and two units on each unit of item x_2 . Assuming that the production quantities cannot be negative, formulate this situation as a linear programming problem.
3. A bakery produces two types of cakes: Chocolate cakes and Vanilla cakes every day. Each Chocolate cake requires 3 kg of flour and 2 hours of baking, while each Vanilla cake requires 2 kg of flour and 1 hour of baking. The bakery has a total of 18 kg of flour and baking time of 8 hours each day. The profit on a Chocolate cake is Rs. 30 and on a Vanilla cake is Rs. 20. Assuming that the bakery wants to maximize the total profit, formulate this situation as a linear programming problem.
4. Consider a knapsack with a maximum capacity W . You are given n items, where each item i has a value v_i and a weight w_i . Formulate the problem of selecting items to be placed in the knapsack such that the total value of the selected items is maximized, while the total weight does not exceed the knapsack capacity. Write the corresponding Linear Programming (LP) formulation of the problem.
5. Now assume that the items are divisible and fractions of items can be selected in the above problem. Formulate the *fractional knapsack problem* and write its corresponding Linear Programming formulation.
6. Use the graphical method to solve the following LP problem.

$$\text{Maximize } Z = 15x_1 + 10x_2$$

$$\begin{aligned}
&\text{subject to} & 4x_1 + 6x_2 &\leq 300, \\
& & x_1 &\leq 60, \\
& & x_2 &\leq 40, \\
& & x_1, x_2 &\geq 0.
\end{aligned}$$

7. Use the graphical method to solve the following LP problem.

$$\begin{aligned}
&\text{Maximize } Z = x_1 + x_2 \\
&\text{subject to} & x_1 - x_2 &\leq 2, \\
& & x_1 + x_2 &\leq 6, \\
& & x_1 - 2x_2 &\leq 1, \\
& & x_1, x_2 &\geq 0.
\end{aligned}$$

8. Solve the LP problems in problems 1, 2 and 3 using the graphical method .

9. Use the graphical method to solve the following LP problems. Also classify whether the following problems has Unique solution, No solution or Infinite solution.

(a)

$$\begin{aligned}
&\text{Minimize } Z = 3x_1 + 2x_2 \\
&\text{subject to} & 5x_1 + x_2 &\geq 10, \\
& & x_1 + x_2 &\geq 6, \\
& & x_1 + 4x_2 &\geq 12, \\
& & x_1, x_2 &\geq 0.
\end{aligned}$$

(b)

$$\begin{aligned}
&\text{Minimize } Z = x_1 + x_2 \\
&\text{subject to} & 3x_1 + 2x_2 &\leq 6, \\
& & 5x_2 + 5x_1 &\geq 0, \\
& & 0 &\leq x_1 \leq 2.5, \\
& & 0 &\leq x_2 \leq 2.
\end{aligned}$$

(c)

$$\begin{aligned}
&\text{Maximize } Z = -x_1 + 2x_2 \\
&\text{subject to} & -x_1 + x_2 &\leq 2, \\
& & 2x_2 - \frac{4}{3}x_1 &\geq 4, \\
& & x_2 &\leq 1 \\
& & x_1 \leq 0, x_2 &\geq 0.
\end{aligned}$$

★ ★ ★ ★ **End** ★ ★ ★ ★